

NE 511: Multi-Physics of Nuclear Reactors

UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

*Lecture 10: Super-prompt Critical Excursion Following Ramp Reactivity
Insertion*

REACTIVITY RAMPS

Note:

- !! Reactivity steps in the super-prompt reactivity domain of reactors are only an idealization.
- !! Realistic simulation of super-prompt-critical reactivity insertions in a reactor must account for the reactivity insertion rate.
- !! In fully realistic situations, ramps are always terminated at some maximum available reactivity.

INVESTIGATION OF THE DIFFERENTIAL EQUATION

The PKA for reactivity ramps is applied only after the time, t_p , when the reactor became super-prompt critical:

$$\dot{p}(\tau) = \frac{\rho_p(\tau)}{\Lambda} p(\tau) \quad \text{with} \quad \tau = t - t_p$$

which is solved with $p(0) = p^0$, and $\rho_p(0) = 0$ for $\tau = 0$

For reactivity ramps the pseudo-initial flux, p^0 , is approximately:

$$p^0 \approx 2p_{pc} \approx p_0 \beta \left(\frac{2\pi}{\Lambda a} \right)^{1/2}$$

or this is approximately two times the actual flux at prompt critical.

INVESTIGATION OF THE DIFFERENTIAL EQUATION

The prompt reactivity with the adiabatic approximation applied to the flux above p_0 is given by

$$\rho_p(\tau) = a\tau + \gamma \int_0^\tau [p(\tau') - p_0] d\tau'$$

or

$$\rho_p(\tau) = a'\tau + \gamma \int_0^\tau [p(\tau')] d\tau' \quad \text{with} \quad a' = a - \gamma p_0$$

The stationary cooling (p_0) only modifies the ramp rate $\rightarrow$ it is included in the feedback reactivity, $\gamma \int_0^\tau [p(\tau') - p_0] d\tau'$.

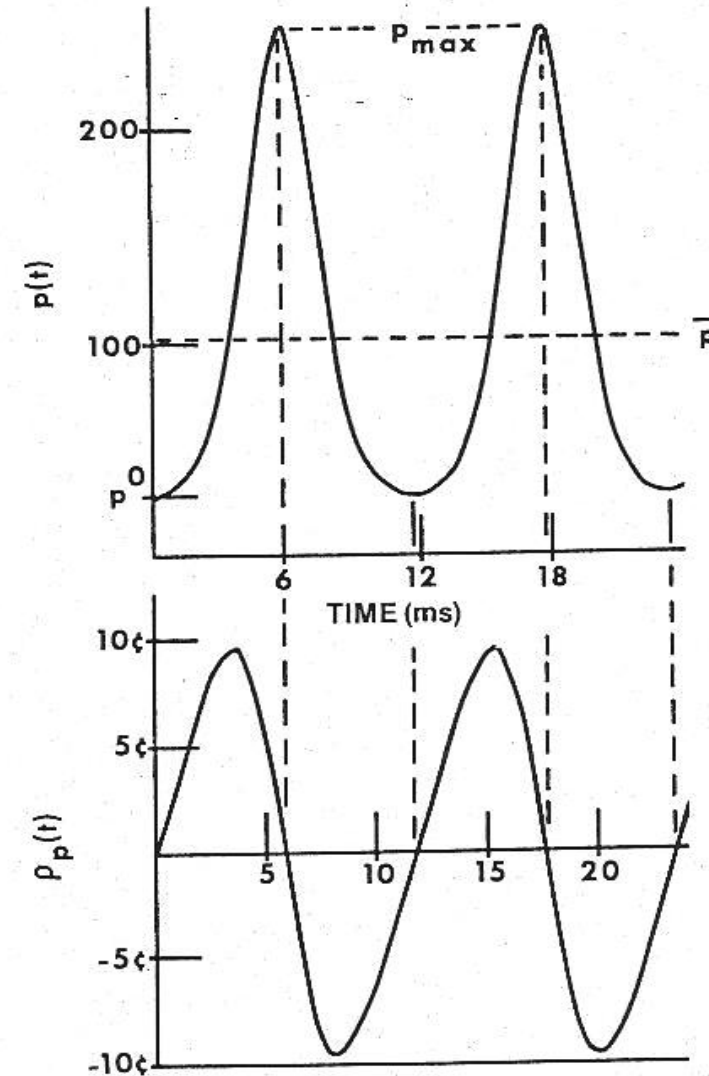
$\dot{p}(\tau)$ and $\rho_p(\tau)$ pass through zero simultaneously:

$$\dot{p}(\tau) = \left(\frac{\rho_p(\tau)}{\Lambda} \right) p(\tau) \rightarrow \dot{p}(\tau) = 0 \quad \text{when} \quad \rho_p(\tau) = 0$$

INVESTIGATION OF THE DIFFERENTIAL EQUATION

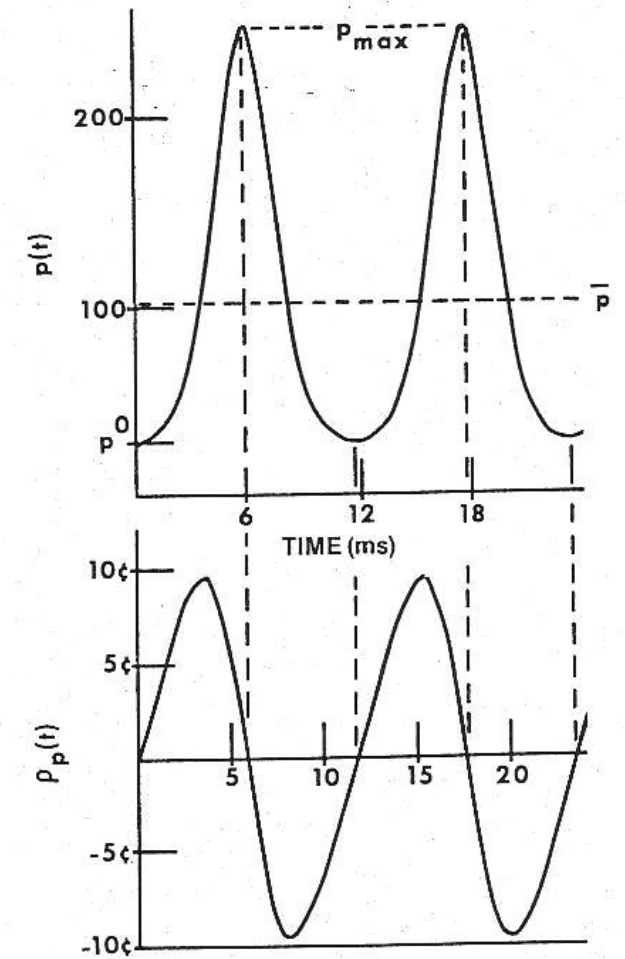
Reactivity-ramp-induced transient with temperature-independent energy coefficient in PKA:

$$p_0 = p_n, a = 50 \frac{\$}{s}, \gamma = -0.5 \frac{\$}{fp-s}, \text{ and } p^0 = 28.0 p_0$$



INVESTIGATION OF THE DIFFERENTIAL EQUATION

- Since $\rho_p(\tau)$ is zero initially, $p(\tau)$ in the PKA starts with a zero slope and has a *minimum* at $\tau = 0$.
- Subsequently, $\rho_p(\tau)$ and $p(\tau)$ both increase.
- With increasing accumulation of energy, the reactivity feedback overcomes the ramp reactivity and $\rho_p(\tau)$ is again reduced to zero; say at $\tau = \tau_{m1}$.
- The flux has its first maximum at this time.
- After passing through zero, ρ_p approaches a negative value, corresponding to ρ_{p1} in the case of a step-induced transient.
- By this time, the high flux during the prompt burst is reduced so that further reactivity reduction through feedback occurs relatively slowly.
- A similar reduction of the reactivity is to be expected in the case of a ramp reactivity insertion.
- However, as the ramp continues to increase the reactivity, it again becomes equal to β .
- At this time, a second flux minimum occurs and the cycle repeats.

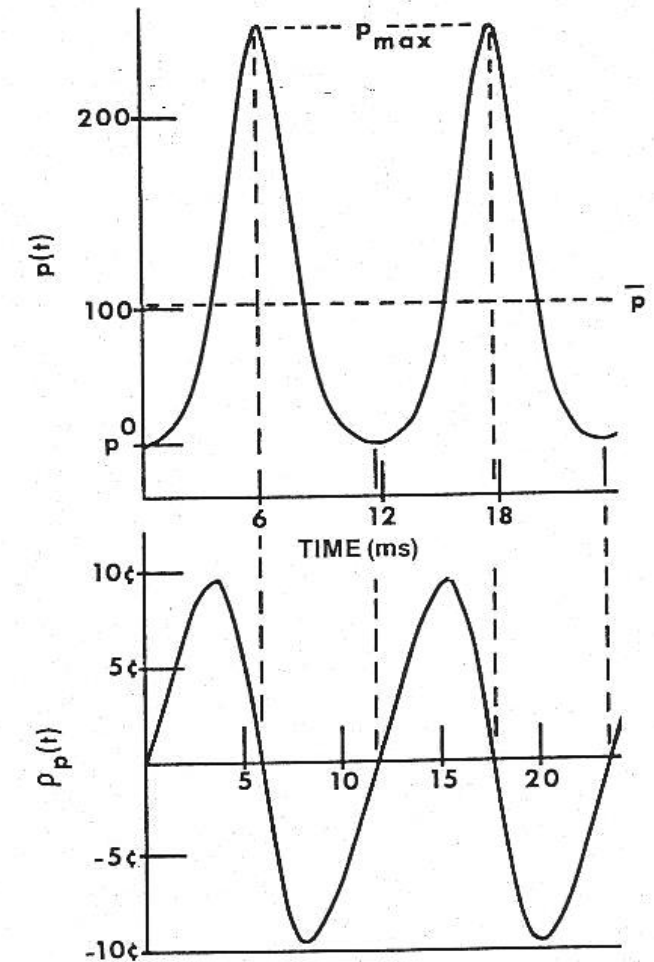


INVESTIGATION OF THE DIFFERENTIAL EQUATION

The discussion of maxima and minima showed that the ramp-induced transient treated in the PKA consists of a repetition of super-prompt flux bursts.

The maxima and minima of the flux occur when $\rho(t)$ passes through β :

$$p(t) = p_{\max} \text{ or } p_{\min} \quad \text{for } \rho_p(t_{ml}) = 0$$



INVESTIGATION OF THE DIFFERENTIAL EQUATION

The succession of bursts can be characterized by an averaged power that can be determined as follows:

1. Let Δt_l be the duration of the l^{th} burst and ΔQ_l the corresponding energy release:

$$\Delta Q_l = \int_{\Delta t_l} [P(t') - P_0] dt'$$

2. Since $\rho_p(t)$ is zero at the beginning and at the end of each burst, the relation between the reactivity insertion and the feedback reactivity **during a burst** can be given by:

$$0 = a\Delta t_l + \gamma_e \int_{\Delta t_l} [P(t') - P_0] dt' = a\Delta t_l + \gamma_e \Delta Q_l$$

INVESTIGATION OF THE DIFFERENTIAL EQUATION

3. The energy release per burst time, i.e., the "average" power increase during the burst, is obtained as:

$$\frac{\Delta Q_l}{\Delta t_l} = \Delta \bar{P} = \bar{P} - P_0 = -\frac{a}{\gamma_e}$$

4. The average power is independent of the generation time and of the initial power → it is the same for all bursts under the conditions of the feedback model.

5. Using an effective ramp rate, the flux amplitude becomes:

$$\bar{p} = -\frac{a'}{\gamma} \quad \text{where} \quad a' = a - \gamma p_0$$

INVESTIGATION OF THE DIFFERENTIAL EQUATION

Note:

For a *step-induced* supercritical transient, information on the *energy release* was obtained by investigating the transition of ρ through β .

For a *ramp-induced* transient, information on the *average power* is obtained instead.

This difference occurs because only the rate of reactivity insertion is given for ramp-induced transients, which then determines the rate of energy release:

$$-\frac{\rho_{p1}}{\gamma_e} \Rightarrow \text{energy (for reactivity steps)}$$

$$-\frac{a}{\gamma_e} \Rightarrow \text{average power (for reactivity ramps)}$$

INVESTIGATION OF THE DIFFERENTIAL EQUATION

Additional information is obtained by **investigating the extrema of $\rho_p(t)$** :

1) setting the time derivative of the prompt reactivity equation equal to zero **and** **2)** applying the adiabatic approximation applied to the energy above p_0 gives:

$$\dot{\rho}_p(\tau) = a + \gamma(p - p_0) = a' + \gamma p = 0.$$

Thus, the flux at the extrema of the reactivity is equal to the average flux :

$$p(\rho_p^{\max, \min}) = p_0 - \frac{a}{\gamma} = -\frac{a'}{\gamma} = \bar{p}$$

i.e., the extrema of the reactivity coincide with $p(t)$ passing through $\bar{p}$.

INVESTIGATION OF THE FIRST INTEGRAL

Re-writing the kinetics equation:

$$\left. \begin{array}{l} \dot{p}(\tau) = \frac{\rho_p(\tau)}{\Lambda} p(\tau) \\ \tau = t - t_p \end{array} \right\} \begin{array}{l} \text{Multiplying by } \dot{p}_p \\ \dot{p}_p(\tau) = a' + \gamma p \\ a' = a + \gamma p_0 \end{array} \left. \right\} \begin{array}{l} (a' + \gamma p) \dot{p} = \frac{1}{\Lambda} \rho_p \dot{p}_p p \\ \underbrace{\hspace{10em}} \\ \text{dividing by } p \\ \left(\frac{a'}{p} + \gamma \right) \dot{p} = \frac{1}{\Lambda} \rho_p \dot{p}_p \end{array}$$

This equation may be readily integrated:

$$\int_0^\tau \left(\frac{a'}{p} + \gamma \right) \dot{p} d\tau' = \frac{1}{\Lambda} \int_0^\tau \rho_p \dot{p}_p d\tau'$$

INVESTIGATION OF THE FIRST INTEGRAL

The derivatives of p and ρ_p may be conveniently combined with $d\tau'$.

Then, the two integrals are carried out over p and ρ_p , respectively:

$$\int_{p(0)}^{p(\tau)} \left(\frac{a'}{p} + \gamma \right) dp = \frac{1}{\Lambda} \int_{\rho_p(0)}^{\rho_p(\tau)} \rho_p d\rho_p$$

Inserting $p(0) = p^0$ and $\rho_p(0) = 0$ as the lower limits gives the "first integral," which represents a relation between the flux and the reactivity:

$$a' \ln \frac{p}{p^0} + \gamma(p - p^0) = \frac{1}{2\Lambda} \rho_p^2$$

INVESTIGATION OF THE FIRST INTEGRAL

The first integral is used to find the maximum and minimum values of both the reactivity and the flux transients:

$$\left. \begin{aligned} a' \ln \frac{p}{p^0} + \gamma(p - p^0) &= \frac{1}{2\Lambda} \rho_p^2 \\ \bar{p} &= -\frac{a'}{\gamma} \end{aligned} \right\} (\rho_p^{\max, \min})^2 = 2\Lambda a' \left(\ln \frac{\bar{p}}{p^0} - 1 + \frac{p^0}{\bar{p}} \right)$$

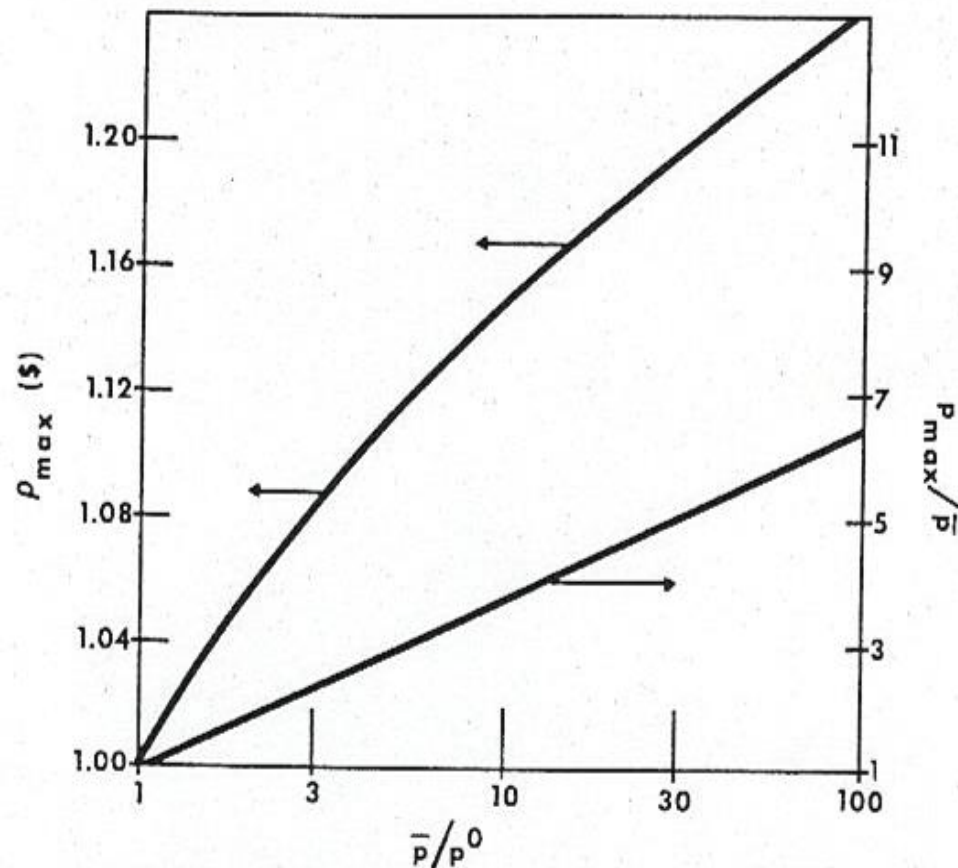
$$\frac{\rho_p^{\max, \min}}{\beta} = \pm \frac{\sqrt{2\Lambda a'}}{\beta} \sqrt{\ln \frac{\bar{p}}{p^0} - 1 + \frac{p^0}{\bar{p}}}$$

1. Positive root represents the reactivity maxima
2. Negative root represents the reactivity minima
3. Maxima and minima of ρ are located symmetrically about β

INVESTIGATION OF THE FIRST INTEGRAL

Example:

Maximum reactivity values versus $\bar{p}/p^0$ for a reactivity-ramp-induced transient where $a = 50\$/s$ and $\gamma = -0.5\$/fp-s$



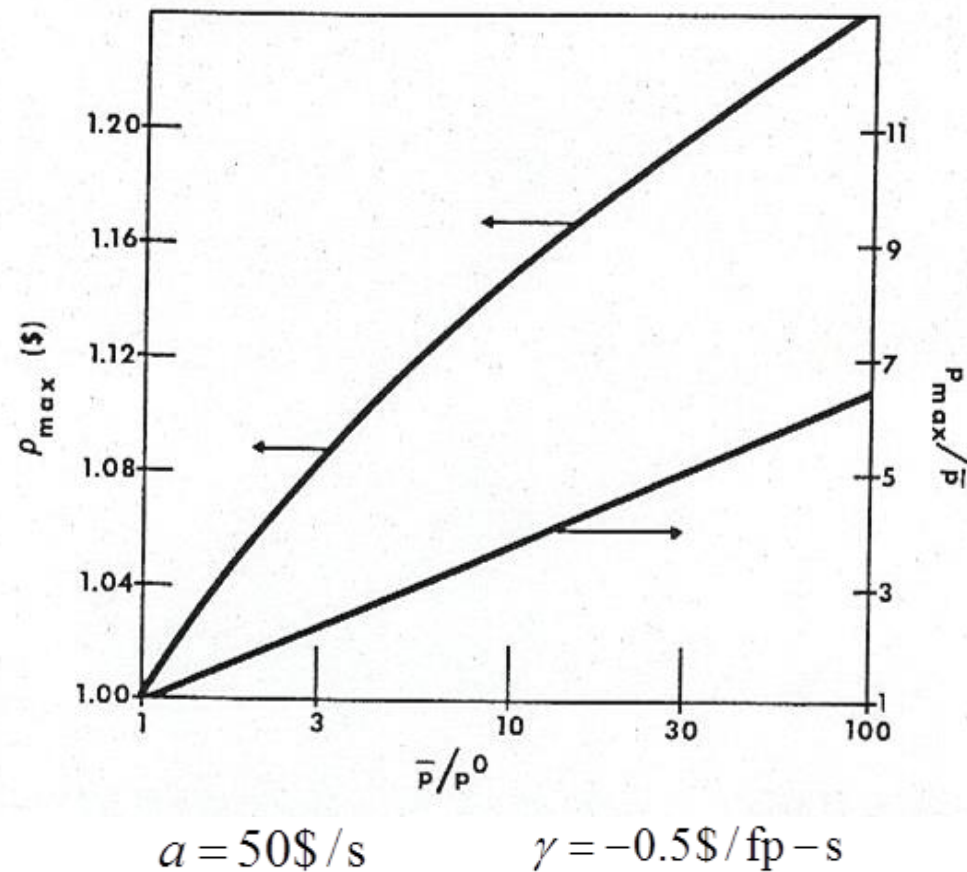
The abscissa covers all transients with realistic ramp rates and feedback coefficients starting at (or around) nominal power

The lower end of the scale is $\bar{p}/p^0 = 1$ since the Equation has real roots only if $\bar{p}/p^0 \geq 1$

If the equation has no real roots, i.e., if $\bar{p} < p^0$ with $\bar{p} = -a'/\gamma$, then the PKA is fundamentally inapplicable

INVESTIGATION OF THE FIRST INTEGRAL

The inequality $\bar{p} < p^0$ may be physically realized for small ramp rates, large feedback coefficients (small $\bar{p}$), or a large pseudo-initial flux (large p^0)



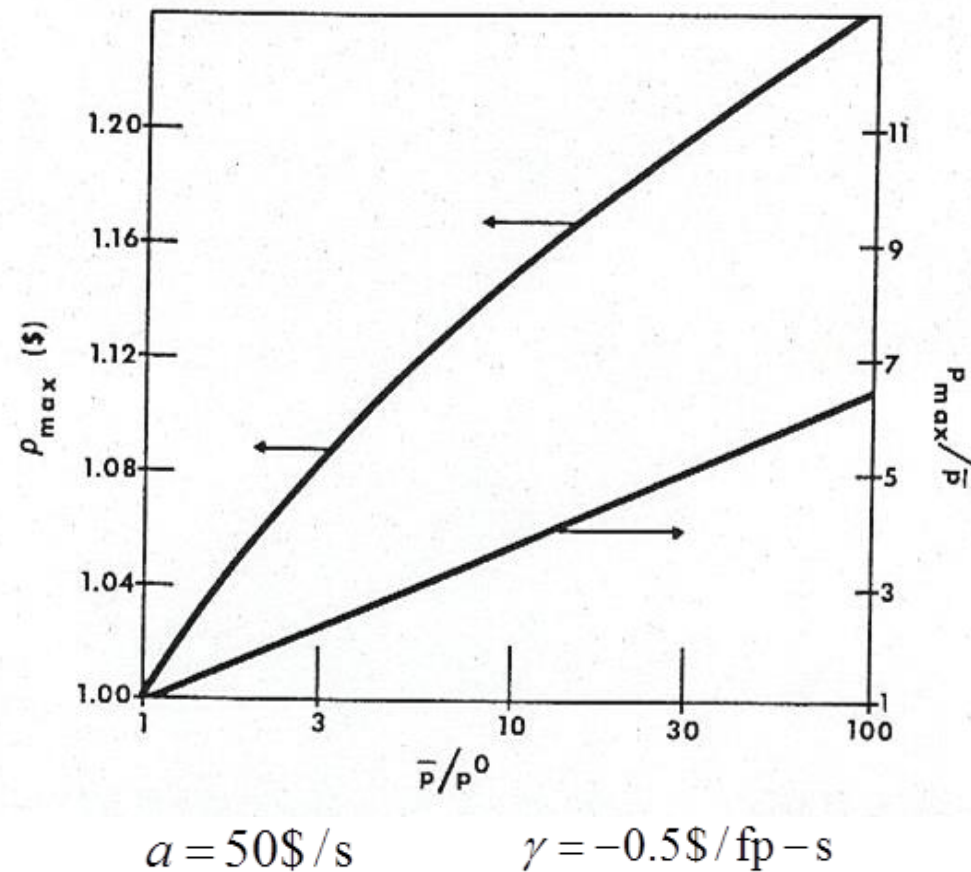
In such cases, the feedback effect during the subprompt-critical part of the transient is strong enough that the reactivity does not reach the superprompt-critical domain

The upper end of the scale in the figure may be exceeded only at very low starting power (e.g., accidents during reloading of the core)

The figure shows that a ramp rate of 50\$/s can only push a fast reactor beyond prompt critical by a small fraction of a dollar

INVESTIGATION OF THE FIRST INTEGRAL

Results for different ramp rates may be obtained readily from this figure by changing the factor in front of the square root



$$\frac{\rho_p^{\max, \min}}{\beta} = \pm \frac{\sqrt{2\Lambda a'}}{\beta} \sqrt{\ln \frac{\bar{p}}{p^0} - 1 + \frac{p^0}{\bar{p}}}$$

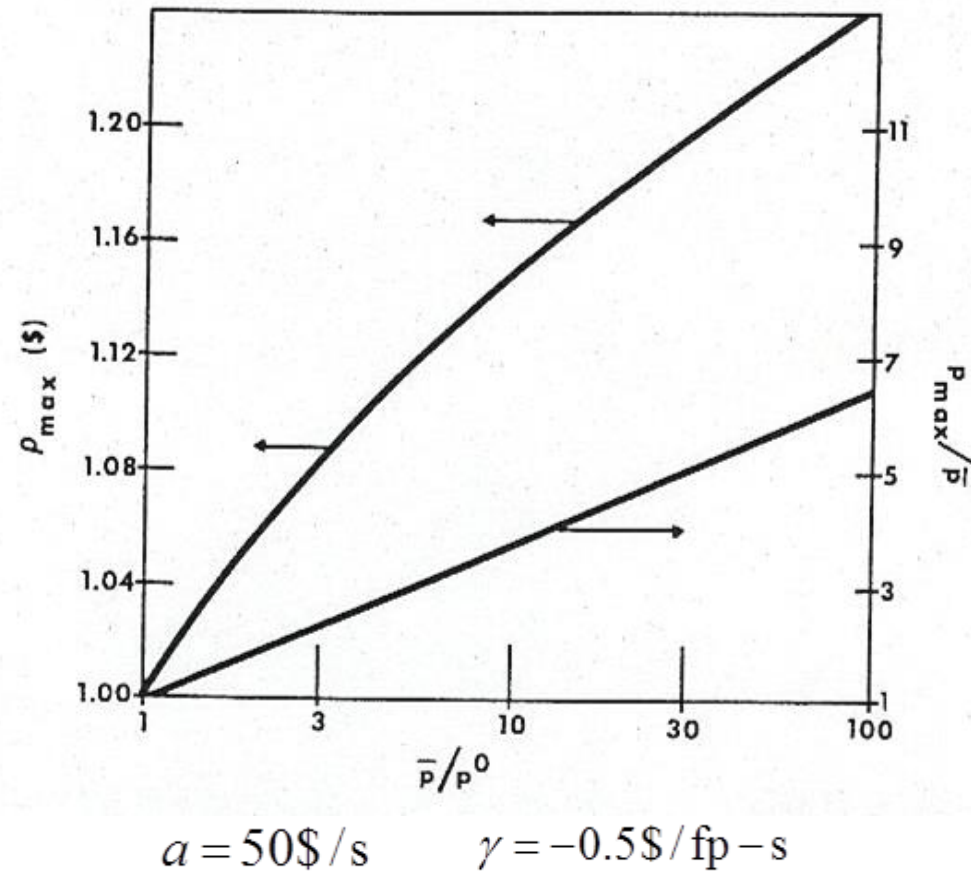
Example:

An increase of the ramp rate from 50 to 100\$/s increases the maximum of ρ_p by a factor of ~ 1.4

INVESTIGATION OF THE FIRST INTEGRAL

Example:

The ratio of the flux maxima to the average flux, $\bar{p}/p^0$, as a function of $p_{max}/\bar{p}$

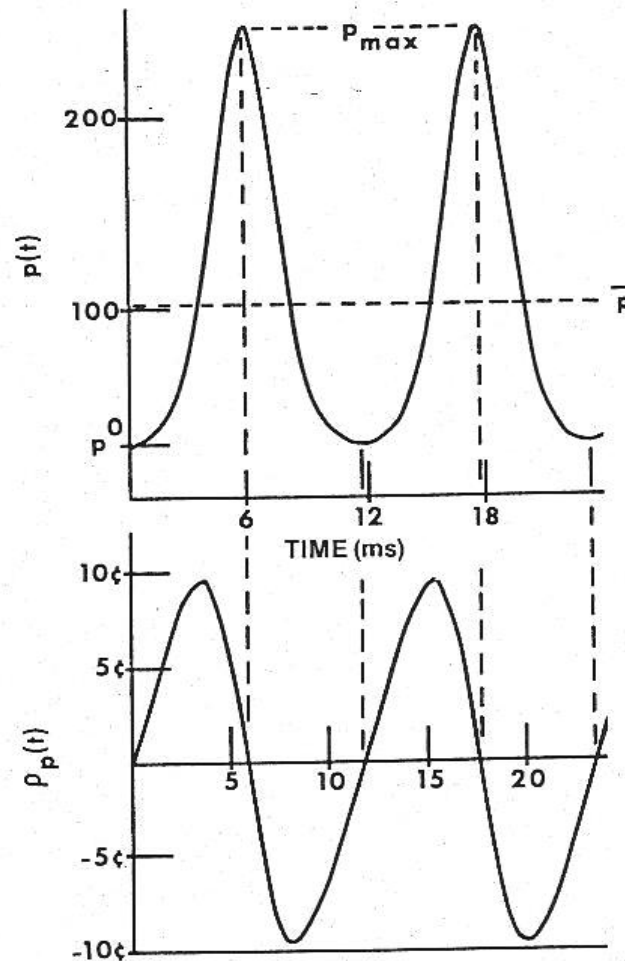


The figure shows that p_{max} does not become very much larger than $\bar{p}$

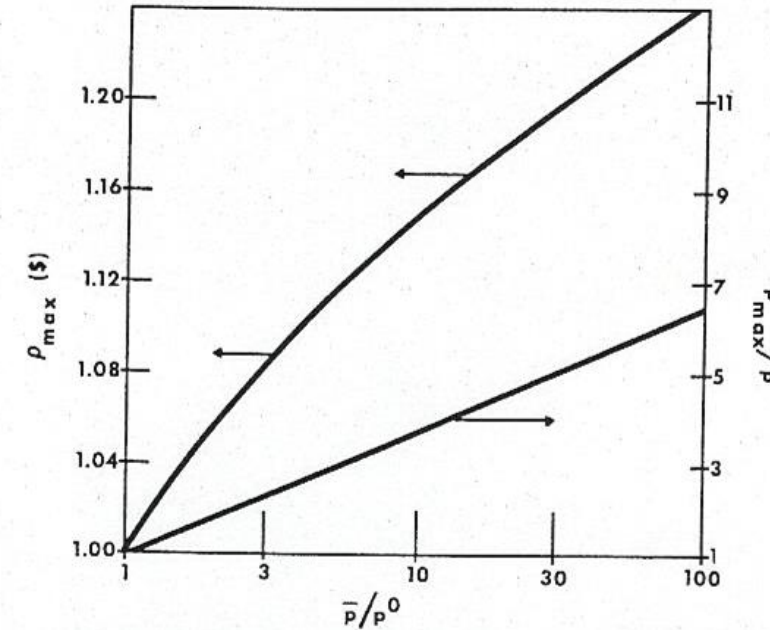
$p_{max}/\bar{p}$ increases with increasing ramp rate since $\bar{p}/p^0$, is approximately proportional to $(a)^{3/2}$

INVESTIGATION OF THE FIRST INTEGRAL

Example 1:



Reactivity-ramp-induced transient with temperature-independent energy coefficient where $p_0 = p_n$, $a = 50\$/s$, $g = -0.5\$/fp - s$, and $p^0 = 28.0 p_0$.



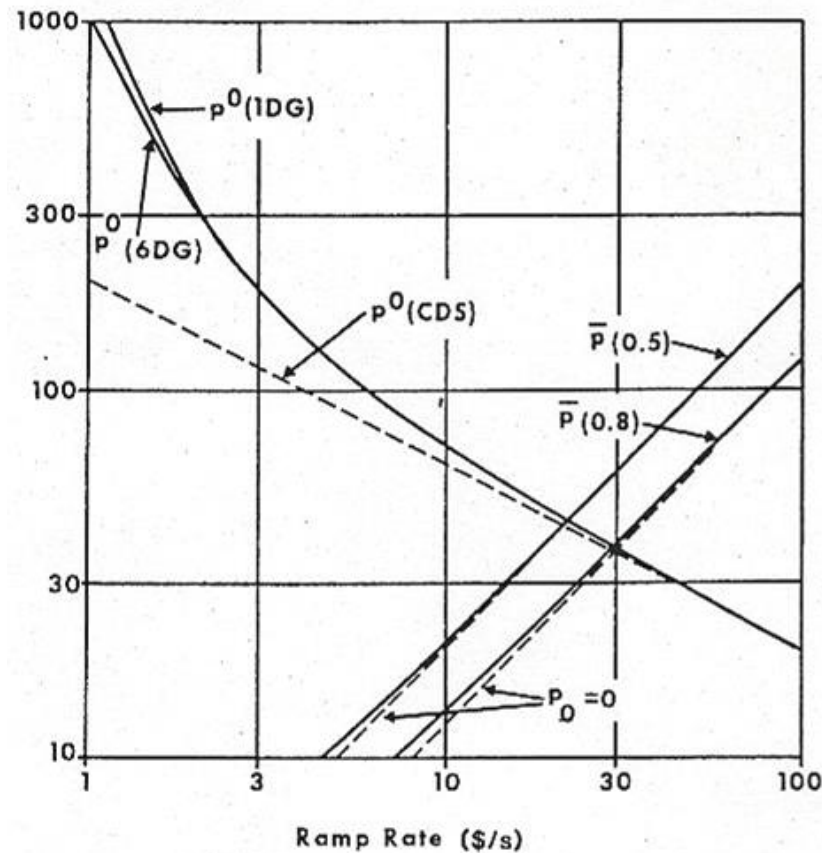
$$\left. \begin{aligned} p_0 &= p_n \\ p^0 &= 28p_0 \\ \bar{p} &= 101p_n \end{aligned} \right\} \bar{p}/p^0 = 3.6$$



$$\frac{p_{max}}{\bar{p}} \approx 2.4 \text{ and } \rho_p^{max} = 9.5\%$$

INVESTIGATION OF THE FIRST INTEGRAL

The investigation of the first integral for ramp-induced transients provided important information such as $\bar{p}$, ρ_{max} , and p_{max}



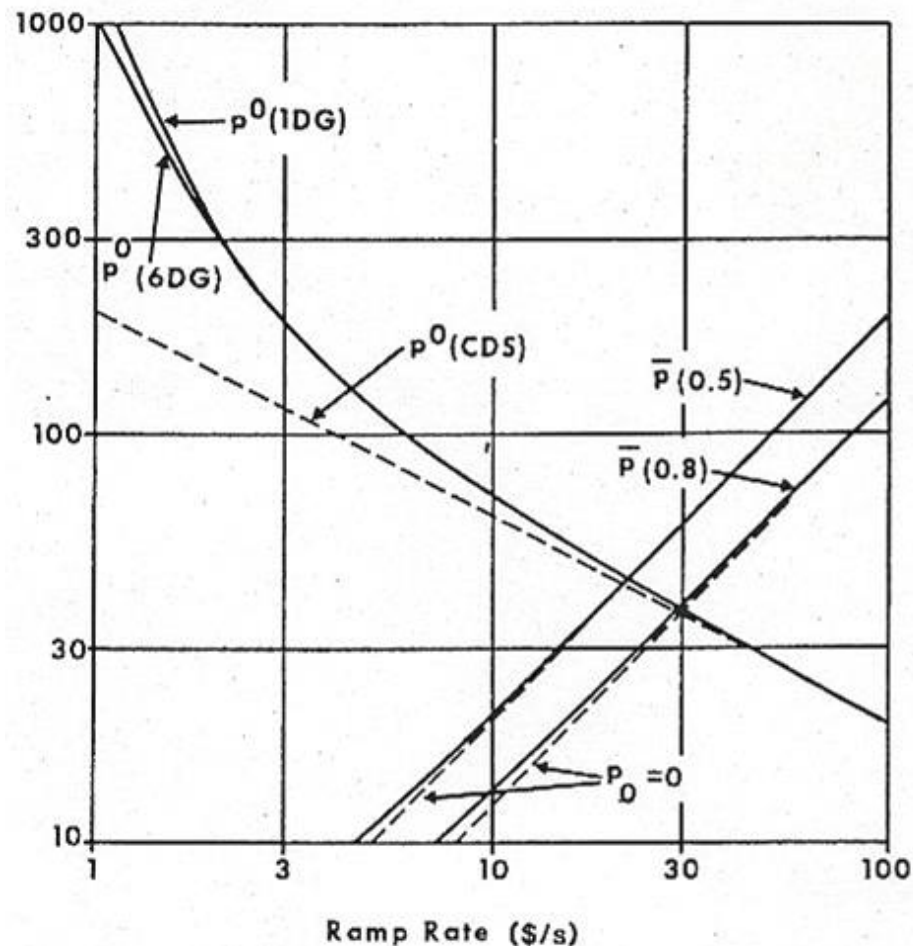
The pseudo-initial flux p^0 decreases and the average flux increases with increasing ramp rate

p^0 and $\bar{p}$ versus reactivity ramp rate ($\gamma = -0.5$ and -0.8 \$/fp-s); the dashed lines are from

$$p^0 = p_0 \beta \sqrt{\frac{2\pi}{\Lambda a}} \quad \bar{p} = -\frac{a'}{\gamma}$$

INVESTIGATION OF THE FIRST INTEGRAL

The PKA can only be applied for ramp rates larger than the crossover point (a^*) of the $p^0(a)$ and curves $\bar{p}(a)$



If the reactivity insertion rate is smaller than a^* , the transient is comparatively slow

Thus, enough time is allowed to accumulate the energy, and thus the feedback reactivity, that is needed to prevent the reactor from becoming superprompt critical

The figure shows that in the domain of applicability of the PKA, the simple formulas for p^0 and $\bar{p}$,

$p^0 = p_0 \beta \sqrt{\frac{2\pi}{\Lambda a}}$ and $\bar{p} = -\frac{a'}{\gamma}$ (see the dashed lines) are sufficiently accurate for semi-quantitative estimates

DISCUSSION OF THE FLUX TRANSIENT

To find the time-dependent flux, the complete solution of the differential equation should be obtained:

$$\dot{p}(\tau) = \frac{\rho_p(\tau)}{\Lambda} p(\tau) \quad \text{with } \tau = t - t_p$$

$$\rho_p(\tau) = a\tau + \gamma \int_0^\tau [p(\tau') - p_0] d\tau'$$

An analytical solution has not been found yet.

- It is possible to calculate the time between the bursts and then deduce the energy per burst from the ratio of the average power and the duration of a burst.
- Therefore, the simple solution of the step-induced transients with a proper ρ_{p1} value is frequently related to ramp-induced transients.

DISCUSSION OF THE FLUX TRANSIENT

So far..

In PKA the delayed neutron source is omitted, and the resulting error is approximately corrected by starting the transient from a pseudo-initial flux.

The value of pseudo-initial flux is determined by matching the PKA solution with a more correct result on the rising wing of the first burst.

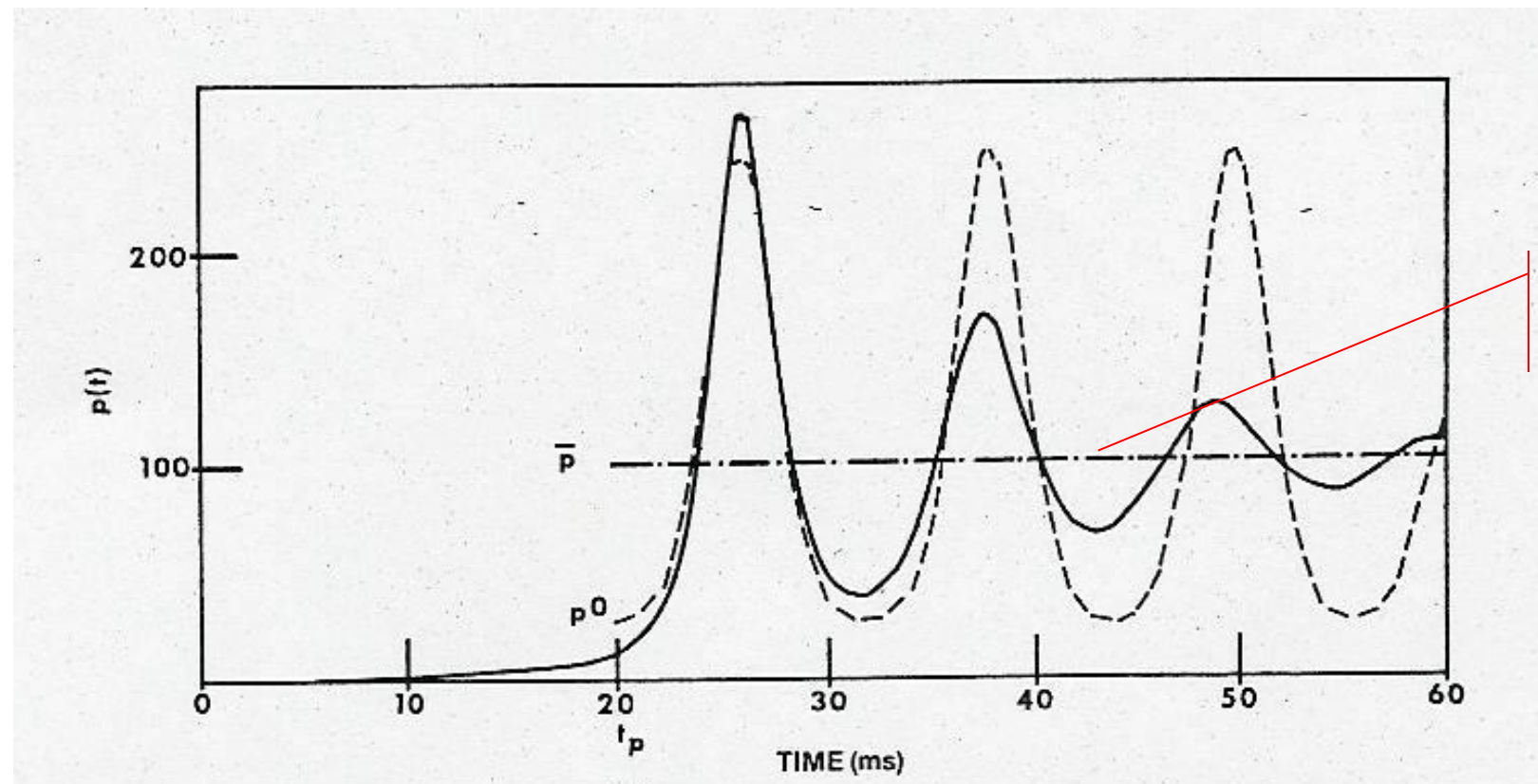
This adjustment yields a fairly accurate PKA solution during the major part of the burst.

The comparison of the PKA for step-induced bursts and the corresponding numerical solution showed that the PKA becomes invalid "after" the burst, i.e., after the flux is reduced to a value comparable to the flux sustained by the delayed neutron source.

DISCUSSION OF THE FLUX TRANSIENT

Reactivity Ramps...

The same phenomenon is to be expected for ramp-induced bursts, i.e., the flux is not expected to decrease to the pseudo-initial flux value.



Effect of Delayed
neutrons

The flux transient induced by a reactivity ramp in the PKA (dashed line) and the actual solution (solid line) with a temperature-independent energy coefficient

Reactivity Ramps...

The Doppler effect contribution to the energy coefficient has approximately a $1/\sqrt{T}$ or a $1/T$ temperature dependence for thermal and fast reactors, respectively.

The temperature increase during the first burst, therefore, causes a reduction in the average energy coefficient effective during the second burst, etc.

Reduction of the energy coefficient from burst to burst has two effects:

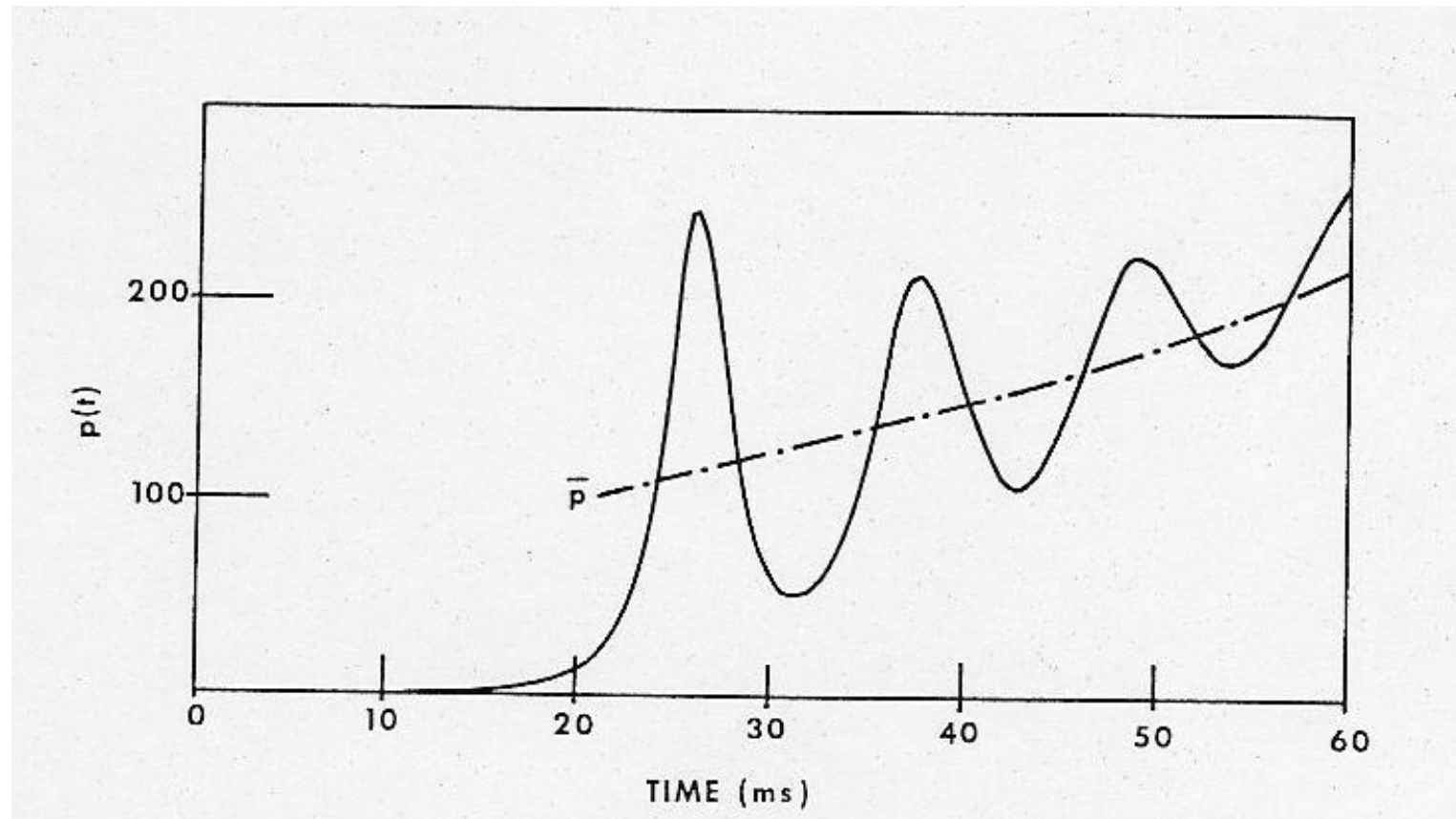
1. It increases $\bar{\rho} \approx -a'/\gamma$
2. It increases the maximum flux

Since $\bar{\rho}$ increases with temperature, it follows that the flux oscillates around a $\bar{\rho}$ curve that increases with time.

DISCUSSION OF THE FLUX TRANSIENT

Reactivity Ramps...

Taking into account the temperature dependence of the energy coefficient:



The flux transient induced by a reactivity ramp - actual solution with a temperature-dependent energy coefficient

DISCUSSION OF THE FLUX TRANSIENT

Reactivity Ramps...

The temperature increase in the constant power removal model is

$$\delta T(t) = T(t) - T_0 = C_{IT} \int_0^t [p(t') - p_0] dt' = C_{IT} I(t)$$

The reactivity feedback is

$$\delta \rho(t) = \int_{T_0}^{T(t)} \frac{A_D}{T} dT = A_D \ln \frac{T(t)}{T_0}$$

or

$$\delta \rho(t) = A_D \ln \left\{ \frac{T_0 + C_{IT} \int_0^t [p(t') - p_0] dt'}{T_0} \right\}$$

Next Class

Delayed Reactivity Feedback Phenomena